GMAT Data Insights

Comprehensive Question Pool

Practice Sets 1-5

100 questions across 5 balanced sets covering all Data Insights question types: Data Sufficiency (DS), Two-Part Analysis (TPA), Graphics Interpretation (GI), Table Analysis (TA), and Multi-Source Reasoning (MSR).

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EDUCATIONAL PURPOSE: This material is intended solely for educational purposes to help students prepare for standardized tests.

1. Question Pool Overview

Pool Structure:

- **Total Questions:** 100 questions across 5 sets
- **Questions per Set:** 20 questions
- **Question Type Distribution per Set:**
  - Data Sufficiency (DS): 8 questions
  - Two-Part Analysis (TPA): 5 questions
  - Graphics Interpretation (GI): 3 questions
  - Table Analysis (TA): 2 questions
  - Multi-Source Reasoning (MSR): 2 questions
- **Difficulty Distribution per Set:**
  - Easy: 6-7 questions
  - Medium: 7-8 questions
  - Hard: 5-6 questions
- **Recommended Time per Set:** 45 minutes

2. Set 1: Mixed Practice

Set 1 Overview:

- DS: 8 questions | TPA: 5 questions | GI: 3 questions | TA: 2 questions | MSR: 2 questions
- Difficulty: 7 Easy, 7 Medium, 6 Hard
- Target Time: 45 minutes

2.1. Data Sufficiency Questions (Set 1)

DS Answer Choices (for all DS questions):
- (A) Statement (1) ALONE is sufficient, but statement (2) alone is not sufficient.
- (B) Statement (2) ALONE is sufficient, but statement (1) alone is not sufficient.
- (C) BOTH statements TOGETHER are sufficient, but NEITHER statement ALONE is sufficient.
- (D) EACH statement ALONE is sufficient.
- (E) Statements (1) and (2) TOGETHER are NOT sufficient.

2.1.1. S1-Q1 (DS - Easy)
Source: DI-GMAT-OG-00274

If a certain vase contains only roses and tulips, how many tulips are there in the vase?

(1) The number of roses in the vase is 4 times the number of tulips in the vase.

(2) There is a total of 20 flowers in the vase.

2.1.2. S1-Q2 (DS - Easy)
Source: DI-GMAT-OG-00277

How many basic units of Currency X are equivalent to 250 basic units of Currency Y?

(1) 100 basic units of Currency X are equivalent to 625 basic units of Currency Y.

(2) 2,000 basic units of Currency X are equivalent to 12,500 basic units of Currency Y.

2.1.3. S1-Q3 (DS - Easy)
Source: DI-GMAT-OG-00278

A company bought 3 printers and 1 scanner. What was the price of the scanner?

(1) The total price of the printers and the scanner was $1,300.

(2) The price of each printer was 4 times the price of the scanner.

2.1.4. S1-Q4 (DS - Medium)
Source: DI-GMAT-OG-00285

If r and s are positive integers, is $\frac{r}{s}$ an integer?

(1) Every factor of s is also a factor of r.

(2) Every prime factor of s is also a prime factor of r.

2.1.5. S1-Q5 (DS - Medium)
Source: DI-GMAT-OG-00290

What is the value of integer n?

(1) $n(n+1) = 6$

(2) $2^{2n} = 16$

2.1.6. S1-Q6 (DS - Hard)

Source: DI-GMAT-OG-00295

In the xy-plane, does the line with equation $y = 3x + 2$ contain the point (r, s)?

(1) $(3r + 2 - s)(4r + 9 - s) = 0$

(2) $(4r - 6 - s)(3r + 2 - s) = 0$

2.1.7. S1-Q7 (DS - Hard)

Source: DI-GMAT-OG-00300

If x is a positive integer greater than 1, is x a prime number?

(1) x has exactly two positive factors.

(2) x is less than 10 and is not divisible by 2 or 3.

2.1.8. S1-Q8 (DS - Hard)

Source: DI-GMAT-SK-00005

There are 210 households in a certain residential complex. All households in the complex that have more than one dog also have at least one cat. All households that have at least one cat have a pet rodent. How many households in the complex have a pet rodent?

(1) 18 households in the complex have two or more dogs.

(2) 90% of households in the complex that have a pet rodent have at least one cat.

2.2. Two-Part Analysis Questions (Set 1)

2.2.1. S1-Q9 (TPA - Easy)

Source: DI-GMAT-OG-00410

A car travels from City A to City B at an average speed of 60 km/h and returns at an average speed of 40 km/h. The distance between the cities is d kilometers.

In terms of d, select the expression that represents the total time for the round trip, and select the expression that represents the average speed for the entire journey.

Total Time	Average Speed	Expression
○	○	$\frac{d}{24}$
○	○	$\frac{d}{48}$
○	○	$\frac{d}{50}$
○	○	48
○	○	50
○	○	$5\frac{d}{120}$

2.2.2. S1-Q10 (TPA - Easy)

Source: DI-GMAT-OG-00412

A store sells two types of widgets: standard and premium. The profit on each standard widget is $5, and the profit on each premium widget is $12. Last month, the store sold a total of 200 widgets.

Select the minimum number of premium widgets that must have been sold for the total profit to be at least $1,500, and select the maximum number of standard widgets that could have been sold for the total profit to be at least $1,500.

Min Premium	Max Standard	Number
○	○	50
○	○	72
○	○	100
○	○	128
○	○	150

2.2.3. S1-Q11 (TPA - Medium)

Source: DI-GMAT-OG-00415

Over a period of 5 academic years from Fall 1999 through Spring 2004, the number of faculty at a certain college increased despite a decrease in student enrollment from 5,500 students in Fall 1999. In the given expressions, F and S represent the percent change in the number of

faculty and students, respectively, over the 5 academic years, and R represents the number of students per faculty member in Fall 1999. The percent change in a quantity X is calculated using the formula $\left(\frac{\{X_{\{\text{new}\}}-X_{\{\text{old}\}}\}}{\{X_{\{\text{old}\}}\}}\right)(100)$.nnSelect the expression that represents the number of faculty in Fall 1999, and select the expression that represents the number of students per faculty member in Spring 2004. Make only two selections, one in each column.

Initial Faculty	Final Ratio	Expression
○	○	$5500R$
○	○	$\frac{5500}{R}$
○	○	$\frac{1}{R}$
○	○	$\left(\frac{100+S}{100+F}\right)R$
○	○	$\left(\frac{100-S}{100+F}\right)R$
○	○	$\left(\frac{100+S}{100-F}\right)R$

2.2.4. S1-Q12 (TPA - Medium)

Source: DI-GMAT-OG-00418

The Quasi JX car has fuel economy E km per liter at constant speed S km per hour under ideal conditions.

Select the expression for liters of fuel used in 1 hour of driving, and select the expression for liters of fuel used in a 60 km drive.

1 Hour	60 km	Expression
○	○	$\frac{S}{E}$
○	○	$\frac{E}{S}$
○	○	$\frac{60}{E}$
○	○	$\frac{60}{S}$
○	○	$\frac{S}{60}$
○	○	$\frac{E}{60}$

2.2.5. S1-Q13 (TPA - Hard)

Source: DI-GMAT-OG-00420

A literature department is organizing a two-day festival featuring ten writers. Five writers will be featured each day. The majority of writers on one day must have a primary writing language that is not English. On the other day, at least four writers must be women. Neither day should have more than two writers from the same country.

Current schedule:

- Day 1: Achebe (male, English, Nigeria), Weil (female, French, France), Gavalda (female, French, France), Barrett Browning (female, English, UK)
- Day 2: Rowling (female, English, UK), Austen (female, English, UK), Ocantos (male, Spanish, Argentina), Lu Xun (male, Chinese, China)

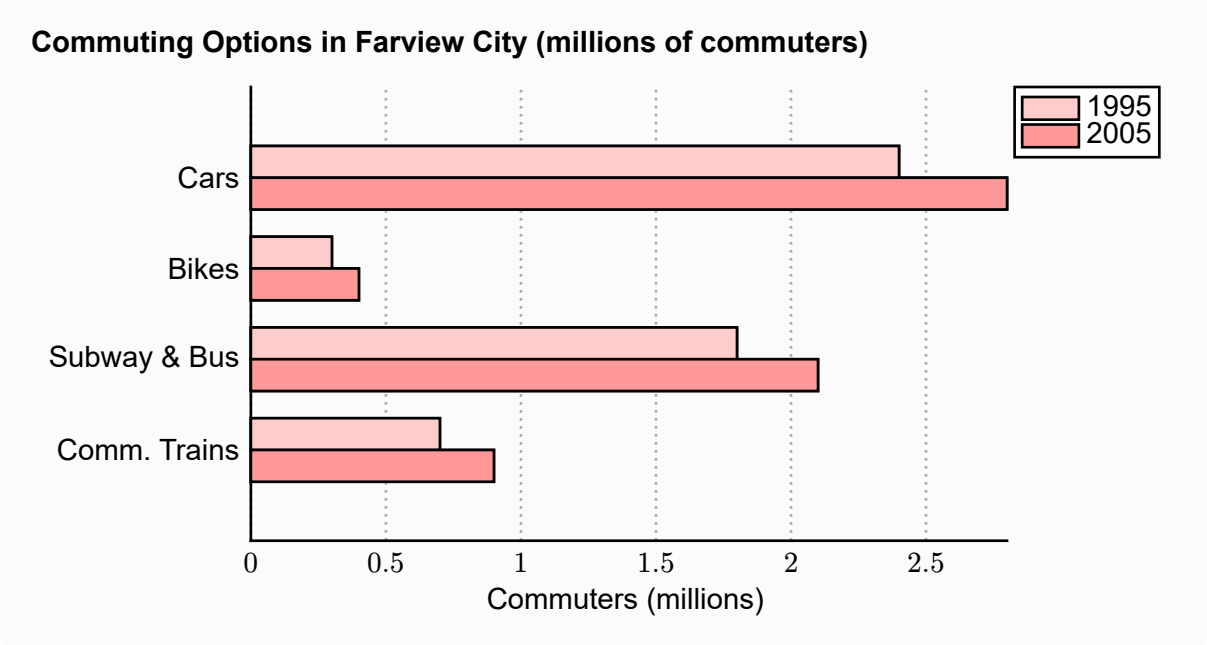
Select a writer who could be added to EITHER day, and select a writer who could be added to NEITHER day.

Either Day	Neither Day	Writer
○	○	LeGuin (female, English, USA)
○	○	Longfellow (male, English, USA)
○	○	Murasaki (female, Japanese, Japan)
○	○	Colette (female, French, France)
○	○	Vargas Llosa (male, Spanish, Peru)
○	○	Zola (male, French, France)

2.3. Graphics Interpretation Questions (Set 1)

2.3.1. S1-Q14 (GI - Easy)

Source: GI-POOL-001



Statement: The commuting mode whose ridership increased by approximately 29% from 1995 to 2005 is [_____]. In 2005, car commuters were approximately [_____] percent of all commuters.

Blank 1 Options:

○ Cars
○ Bikes
○ Subway & Bus
○ Commuter Trains

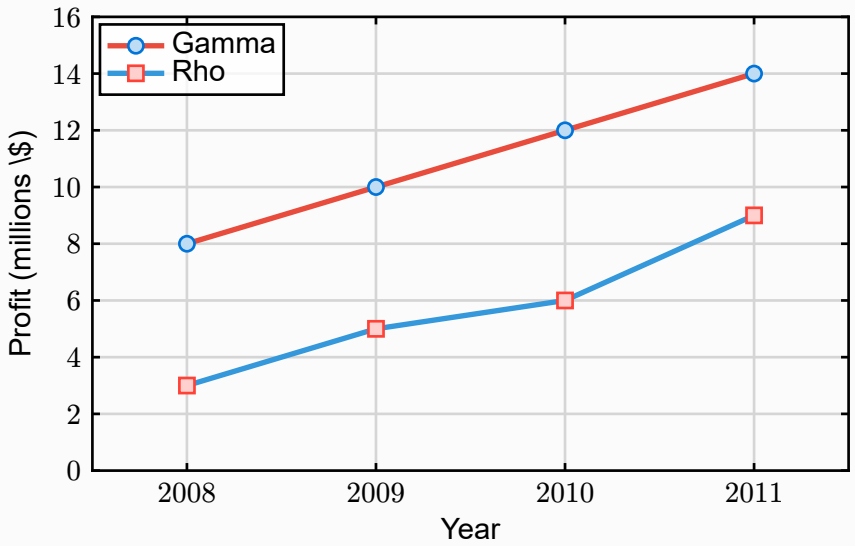
Blank 2 Options:

○ 28%
○ 32%
○ 45%
○ 52%

2.3.2. S1-Q15 (GI - Medium)

Source: GI-POOL-002

Division Profits 2008-2011 (millions $)



Statement: In 2010 and 2011 combined, Rho division accounted for ____ percent of the company's total profits from both divisions. From 2008 to 2011, Rho division's profits increased by ____ percent.

Blank 1 Options:

○ 26.8%
○ 36.6%
○ 42.4%
○ 57.6%

Blank 2 Options:

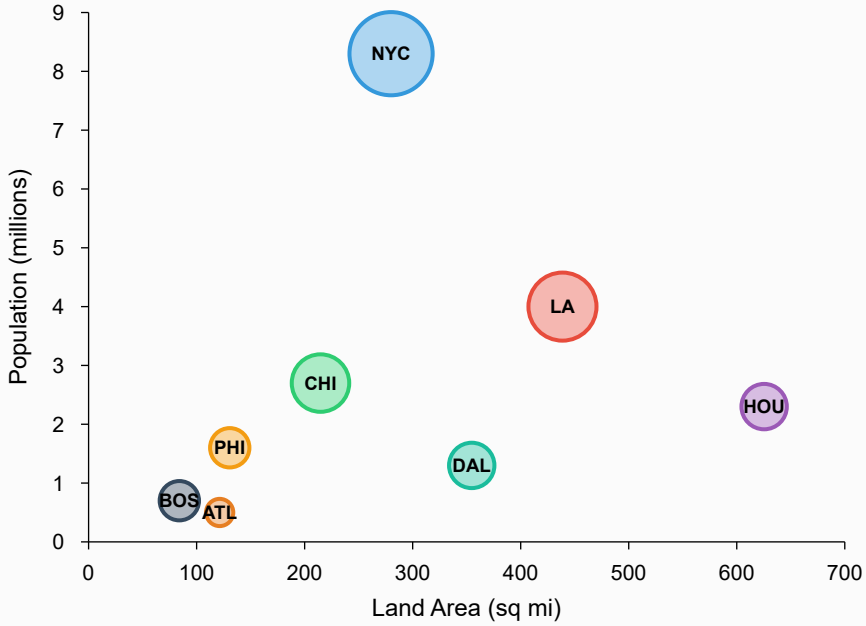
○ 100%
○ 150%
○ 200%
○ 300%

2.3.3. S1-Q16 (GI - Hard)

Source: GI-POOL-003

US Cities: Population vs. Land Area

Bubble size represents GDP of metropolitan region



Statement: Among the eight cities shown, ____ is the city with the lowest population density.

According to this graph, the population of a city is ____ with the size of the GDP of its metropolitan region.

Blank 1 Options:

○ Atlanta
○ Houston
○ Dallas
○ Boston

Blank 2 Options:

○ positively correlated
○ negatively correlated
○ not correlated

2.4. Table Analysis Questions (Set 1)

2.4.1. S1-Q17 (TA - Medium)

Source: TA-POOL-001

The table shows tertiary education data for selected countries:

Country	% of pop. over 20 in tertiary ed.	Public spending per student ($)
Sweden	42%	53.50
Finland	38%	48.20
Denmark	35%	51.00
Norway	32%	49.80
Australia	28%	42.30
UK	25%	38.50
Germany	22%	45.60
France	18%	36.20
Italy	15%	28.40
Spain	12%	31.50

For each statement, select **Yes** if it can be inferred, or **No** if it cannot.

Statement	Yes	No
No country with more than 25% of people over 20 in tertiary programs spends more than $53.50 per student.	○	○
Every country with more than 30% enrollment spends at least $48 per student.	○	○
The median spending per student for these 10 countries is between $40 and $45.	○	○

2.4.2. S1-Q18 (TA - Hard)

Source: TA-POOL-007

The table shows quarterly revenue data for five regional offices of a consulting firm (in $millions):

Office	Q1	Q2	Q3	Q4	Total
Boston	4.2	5.1	4.8	6.3	20.4
Chicago	3.8	4.2	5.5	5.8	19.3
Denver	2.1	2.8	3.2	3.6	11.7
Miami	3.5	4.0	3.8	4.5	15.8
Seattle	2.9	3.3	4.1	4.8	15.1

Note: Each office has a different number of consultants. Boston has the most consultants and Denver has the fewest.

For each statement, select **Must be true** or **Need not be true**.

Statement	Must be true	Need not be true
Boston had the highest revenue per consultant in Q4.	○	○
Denver had the lowest revenue per consultant for the year.	○	○
Every office had higher revenue in Q4 than in Q1.	○	○

2.5. Multi-Source Reasoning Questions (Set 1)

2.5.1. S1-Q19 & S1-Q20 (MSR - Medium)

Source: MSR-POOL-001

Tab 1 — Email from Administrator

From: Sarah Chen, Project Administrator
To: Research Team
Subject: Marketing Survey Response Rate
Date: March 15

Team, we need 700 completed surveys for statistical significance. Our budget allows $15 per completed survey, with $12,000 allocated for respondent compensation. Please update me on current status.

Tab 2 — Email from Coordinator

From: Marcus Webb, Project Coordinator
To: Sarah Chen
Date: March 16

Sarah, as of this morning we have 350 completed surveys. Based on current rates, we project 50 additional responses per day. We have 10 days until the planned close date. Note that response rates typically decline in final days.

Tab 3 — Budget Summary

Item	Amount
Total Budget	$15,000
Platform Fees	$2,000
Respondent Compensation	$12,000
Contingency Fund	$1,000

Note: We have agreed to try not to exceed the allocated budget.

S1-Q19: For each statement, select **Yes** or **No**.

Statement	Yes	No
The survey has currently received enough responses to meet the target.	○	○
Based on projections, the survey will definitely reach 700 responses before close.	○	○
It is possible the project will need to exceed the compensation budget.	○	○

S1-Q20: Based on the information provided, what is the maximum number of surveys that can be completed within the allocated respondent compensation budget?

◯ 700
◯ 750
◯ 800
◯ 850
◯ 1000

3. Set 1: Answer Key

Complete all questions before reviewing answers!

3.1. DS Answers (Set 1)

Q	Answer	Difficulty	Key Insight
S1-Q1	C	Easy	Ratio 4:1 + total 20 → tulips = 4
S1-Q2	D	Easy	Each gives exchange rate to find 250Y in X
S1-Q3	C	Easy	3P + S = 1300 and P = 4S → 13S = 1300 → S = 100
S1-Q4	A	Medium	Statement 1 ensures divisibility; Statement 2 is insufficient
S1-Q5	D	Medium	Both give n = 2
S1-Q6	C	Hard	Need both to ensure (r,s) satisfies y = 3x + 2
S1-Q7	A	Hard	Statement 1 is definition of prime; Statement 2 allows 5 or 7
S1-Q8	E	Hard	Neither statement alone nor together determines rodent count

3.2. TPA Answers (Set 1)

- **S1-Q9:** Total Time = $5\frac{d}{120}$, Average Speed = 48
- **S1-Q10:** Min Premium = 72, Max Standard = 128
- **S1-Q11:** Initial Faculty = $\frac{5500}{R}$, Final Ratio = $\left(\frac{100+S}{100+F}\right)R$
- **S1-Q12:** 1 Hour = $\frac{S}{E}$, 60 km = $\frac{60}{E}$
- **S1-Q13:** Either Day = Murasaki, Neither Day = Longfellow

3.3. GI Answers (Set 1)

- **S1-Q14:** Blank 1 = Commuter Trains (0.7→0.9 = 28.6%), Blank 2 = 45% (2.8/6.2)
- **S1-Q15:** Blank 1 = 36.6% (15/41), Blank 2 = 200% ((9-3)/3)
- **S1-Q16:** Blank 1 = Houston (lowest density), Blank 2 = positively correlated

3.4. TA Answers (Set 1)

- **S1-Q17:** Yes, Yes, Yes
- **S1-Q18:** Need not be true, Need not be true, Must be true (all offices increased Q1→Q4)

3.5. MSR Answers (Set 1)

- **S1-Q19:** No, No, Yes
- **S1-Q20:** 800 ($12,000 ÷ $15 = 800)

4. Set 2: Mixed Practice

Set 2 Overview:

- DS: 8 questions | TPA: 5 questions | GI: 3 questions | TA: 2 questions | MSR: 2 questions
- Difficulty: 7 Easy, 7 Medium, 6 Hard
- Target Time: 45 minutes

4.1. Data Sufficiency Questions (Set 2)

4.1.1. S2-Q1 (DS - Easy)

Source: DI-GMAT-OG-00310

What is the value of x?

(1) $x + 1 = 2$

(2) $(x + 1)^2 = 4$

4.1.2. S2-Q2 (DS - Easy)

Source: DI-GMAT-OG-00315

Is the integer n odd?

(1) n is divisible by 3.

(2) n is divisible by 5.

4.1.3. S2-Q3 (DS - Easy)

Source: DI-GMAT-OG-00320

If x and y are positive integers, what is the value of $x + y$?

(1) $xy = 6$

(2) $x < y$

4.1.4. S2-Q4 (DS - Medium)

Source: DI-GMAT-OG-00325

If a and b are integers, is $a + b$ divisible by 7?

(1) a is divisible by 7.

(2) b is divisible by 7.

4.1.5. S2-Q5 (DS - Medium)

Source: DI-GMAT-OG-00330

A group of 8 machines produces widgets. What is the average number of widgets produced per machine?

(1) The total number of widgets produced is 320.

(2) The machines produced widgets at the same rate.

4.1.6. S2-Q6 (DS - Medium)

Source: DI-GMAT-OG-00335

Is $x > y$?

(1) $x = y + 2$

(2) $\frac{x}{2} = \frac{y}{2} + 1$

4.1.7. S2-Q7 (DS - Hard)

Source: DI-GMAT-OG-00340

If x, y, and z are positive numbers, is $x > y > z$?

(1) $xz > yz$

(2) $\frac{y}{x} < 1$

4.1.8. S2-Q8 (DS - Hard)

Source: DI-GMAT-OG-00345

In a certain sequence, each term after the first is found by multiplying the previous term by a constant. If the 3rd term is 24 and the 6th term is 192, what is the 1st term?

(1) The 4th term is 48.

(2) The common ratio is positive.

4.2. Two-Part Analysis Questions (Set 2)

4.2.1. S2-Q9 (TPA - Easy)

Source: DI-GMAT-PQ-00114

A store's revenue from selling n items at price p each is $R = np$. If the store sells 150 items and wants revenue of at least $3,000:

Select the minimum price per item, and select the actual revenue if items are sold at $25 each.

Min Price	Revenue at $25	Value
○	○	$15
○	○	$20
○	○	$25
○	○	$3,000
○	○	$3,750
○	○	$4,500

4.2.2. S2-Q10 (TPA - Easy)

Source: DI-GMAT-PQ-00116

A tank contains 200 liters of a 30% salt solution. Pure water is added to dilute the solution.

Select the amount of water needed to create a 20% solution, and select the final volume of the 20% solution.

Water Added	Final Volume	Liters
○	○	50
○	○	100
○	○	200
○	○	250
○	○	300

4.2.3. S2-Q11 (TPA - Medium)

Source: DI-GMAT-PQ-00120

Two trains leave stations 480 km apart, traveling toward each other. Train A travels at 80 km/h and Train B at 60 km/h.

Select the time until they meet, and select the distance traveled by Train A when they meet.

Time	Distance (A)	Value
○	○	2.5 hours

○	○	3 hours
○	○	3.43 hours
○	○	200 km
○	○	240 km
○	○	274 km

4.2.4. S2-Q12 (TPA - Medium)

Source: DI-GMAT-PQ-00125

A company's profit function is $P(x) = -2x^2 + 120x - 1000$, where x is units sold in thousands.

Select the number of units that maximizes profit, and select the maximum profit.

Units (thousands)	Max Profit	Value
○	○	20
○	○	30
○	○	60
○	○	$600
○	○	$800
○	○	$1,000

4.2.5. S2-Q13 (TPA - Hard)

Source: DI-GMAT-PQ-00130

A committee of 5 must be formed from 6 men and 4 women. The committee must have at least 2 women.

Select the number of committees with exactly 2 women, and select the number of committees with at least 2 women.

Exactly 2 Women	At Least 2 Women	Number
○	○	60
○	○	120
○	○	186
○	○	246
○	○	252

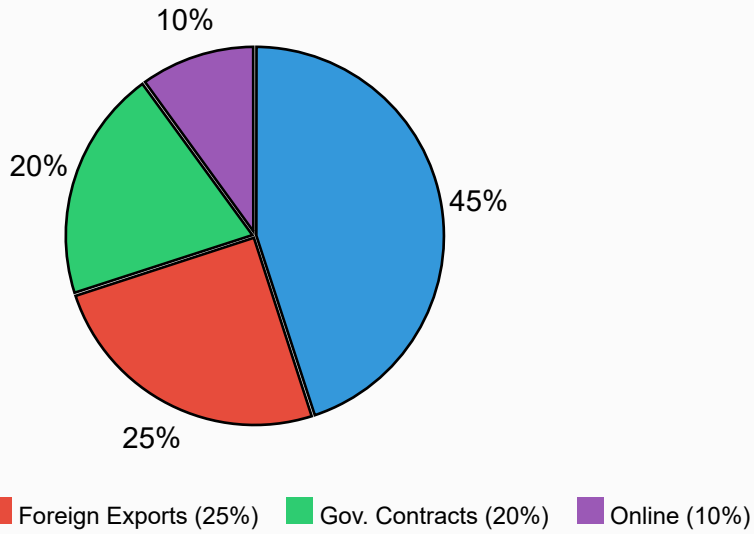
4.3. Graphics Interpretation Questions (Set 2)

4.3.1. S2-Q14 (GI - Easy)

Source: GI-POOL-004

Goliath Corporation's 2010 Revenue Distribution

Total revenue: $500 million



Grocery breakdown by region: Northeast 30%, Midwest 40%, South 20%, West 10%

Statement: The revenue from foreign exports is ____ the revenue from grocery stores in the Northeast. Revenue from government contracts would have to increase by approximately ____ to equal revenue from grocery sales in the Midwest.

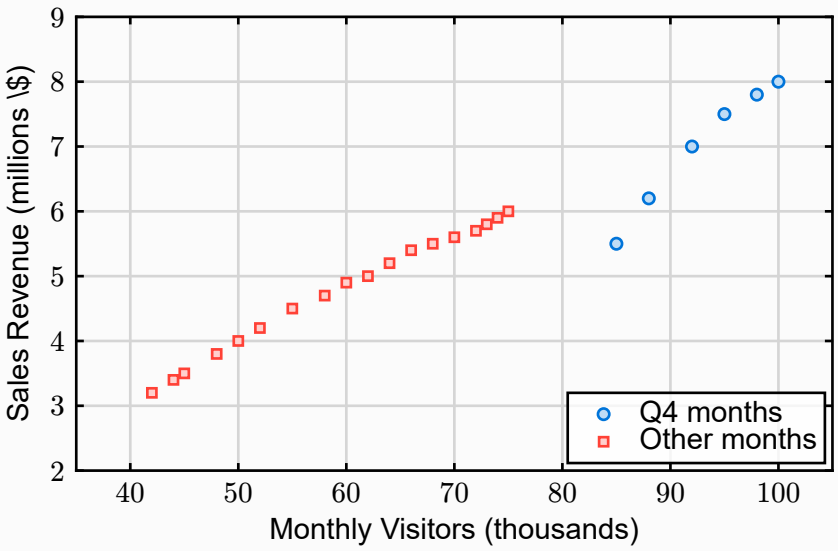
Blank 1 Options:
○ greater than
○ less than
○ equal to

Blank 2 Options:
○ 50%
○ 80%
○ 101%
○ 125%

4.3.2. S2-Q15 (GI - Medium)

Source: GI-POOL-005

Apex Appliance Monthly Data (2011-2012)



Statement: During this two-year period, Apex had ____ non-fourth-quarter month(s) with higher sales revenue than the fourth-quarter month with the lowest sales. In the month with the highest sales yield (revenue per visitor), Apex earned approximately ____ in revenue.

Blank 1 Options:

○ 0
○ 1
○ 2
○ 3

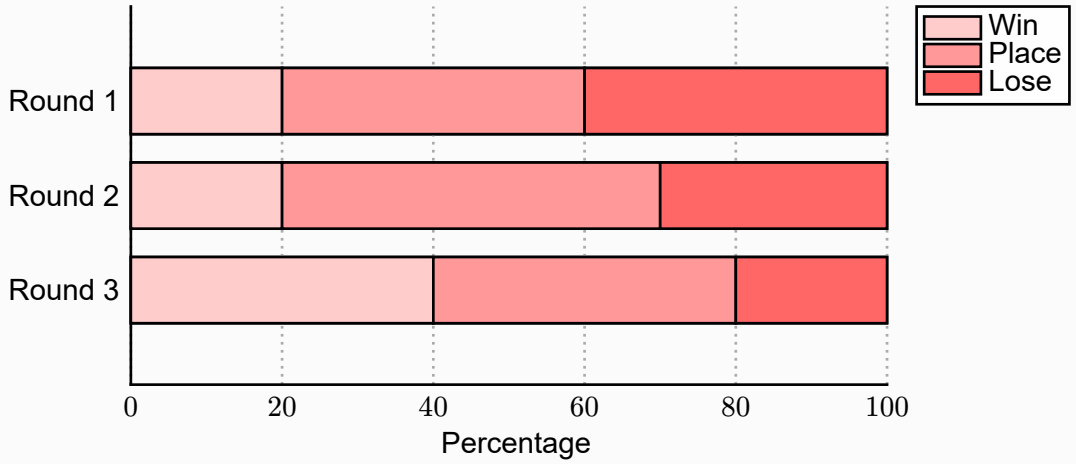
Blank 2 Options:

○ \$5.5 million
○ \$6.5 million
○ \$7.5 million
○ \$8.0 million

4.3.3. S2-Q16 (GI - Hard)

Source: GI-POOL-006

Academic Competition Results by Round



Only those who Win or Place advance to the next round. Commendations are awarded for each Win or Place.

Statement: If 100,000 participants start, ____ of them would win the entire competition. Exactly ____ % of participants who start acquire exactly two commendations.

Blank 1 Options:

○ 800
○ 2,400
○ 4,800
○ 8,000

Blank 2 Options:

○ 14.4%
○ 21.6%
○ 28.8%
○ 36.0%

4.4. Table Analysis Questions (Set 2)

4.4.1. S2-Q17 (TA - Easy)

Source: TA-POOL-003

Brazilian agricultural products in 2009:

Commodity	Prod. Share	Prod. Rank	Export Share	Export Rank
Coffee	40%	1	32%	1
Orange juice	56%	1	82%	1
Sugar	21%	1	44%	1
Soybeans	27%	2	40%	2
Beef	16%	2	22%	1
Chickens	15%	3	38%	1
Corn	8%	4	10%	2
Cotton	5%	5	10%	4
Pork	4%	4	12%	4

For each statement, select **Yes** or **No**.

Statement	Yes	No
Brazil ranks first in both production and exports for exactly 3 commodities.	○	○
For every commodity where Brazil ranks first in exports, Brazil produces at least 15% of world supply.	○	○
The median export share for these commodities is between 20% and 30%.	○	○

4.4.2. S2-Q18 (TA - Medium)

Source: TA-POOL-008

Product sales data for an electronics retailer (units sold by category):

Product	Jan	Feb	Mar	Avg Price
Laptops	120	145	132	$850
Tablets	85	92	110	$420
Phones	210	195	240	$680
Monitors	65	78	71	$320
Accessories	340	285	310	$45

Note: Revenue = Units × Price

For each statement, select **Yes** if it can be determined or **No** if it cannot.

Statement	Yes	No
The category with highest Q1 revenue was Phones.	○	○
Total Q1 revenue from all categories exceeded $500,000.	○	○
Tablets had the largest percentage increase in units from Jan to Mar.	○	○

4.5. Multi-Source Reasoning Questions (Set 2)

4.5.1. S2-Q19 & S2-Q20 (MSR - Hard)

Source: MSR-POOL-005

Tab 1 — Greenhouse Gas Reduction Policy

The city council is evaluating proposals to reduce greenhouse gas emissions. Each proposal has an implementation cost, estimated annual savings, and projected emission reduction. A proposal must achieve at least 15% emission reduction to qualify for federal matching funds.

Tab 2 — Proposal Details

Proposal	Cost (M)	Annual Savings	Emission Cut
Solar Panel Initiative	$12	$2.1M	18%
Public Transit Expansion	$25	$4.5M	22%
Building Efficiency	$8	$1.8M	12%
Electric Fleet Conversion	$15	$2.8M	16%
Bike Infrastructure	$5	$0.6M	8%

Tab 3 — Budget Constraints

The city has $30 million available. Federal matching would cover 50% of qualifying projects' costs. The council prefers projects with payback periods under 6 years (Payback = Cost / Annual Savings).

S2-Q19: For each statement, select **Yes** or **No**.

Statement	Yes	No
Building Efficiency qualifies for federal matching funds.	○	○
The city can afford all three qualifying proposals without federal help.	○	○
Public Transit Expansion has the shortest payback period of the qualifying proposals.	○	○

S2-Q20: If the city chooses only proposals that qualify for federal matching and have payback periods under 6 years, what is the maximum emission reduction achievable?

○ 18%
○ 34%
○ 40%
○ 56%